PowerMCU

檔案： PowerMCU.kicad_sch

Measurement

檔案： Measurement.kicad_sch

DisplayUI

檔案： DisplayUI.kicad_sch

HarvestPower

檔案： HarvestPower.kicad_sch

HarvestSwitch

檔案： HarvestSwitch.kicad_sch

AlwaysOn

檔案： AlwaysOn.kicad_sch

ADG706 harvest+measurement schematic candidate / NOT FOR FABRICATION

DSSC -> AEM10941 -> two supercapacitors -> TPS61023 -> LOLIN VBUS

DSSC -> relay -> 16 load candidates -> ADG706 -> GND; ADS1115 senses DUT.

Global net labels connect sheets. All parts instantiated; actual pins shown on child sheets.

Harvest/relay circuit under integration; no routing or fabrication release.

Review gate: verify maximum illuminated Voc/Isc, relay contacts, firmware GPIO/resistor table and physical assembly.

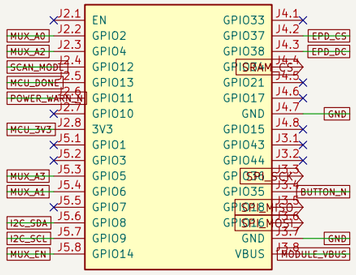
PCB is mechanical baseline + interfaces + off-board staging; NO SIGNAL ROUTING.

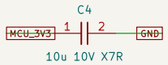
Based on Marc Dusseiller IVcurve_tester f9a75be; new integrated design

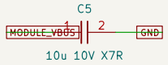
Sheet: /
File: Marc_USB.kicad_sch

Title: Marc USB – Integrated draft

Size: A4	Date: 2026-10-18	**Rev: D0**
KiCad E.D.A. 10.0.5		Id: 1/7









USB ONLY: use module USB-C; no second carrier USB supply and no external 3V3 input.

VBUS and MCU_3V3 are module outputs. EPD 3V3_OUT must NEVER join MCU_3V3.

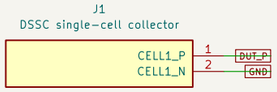
Onboard ME6211C33: current/thermal budget and recessed USB plug fit NOT qualified.

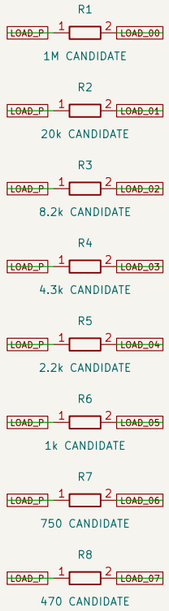
J2/J5/J4/J3 pin names match WEMOS V1.0.0 schematic. Antenna / height review pending.

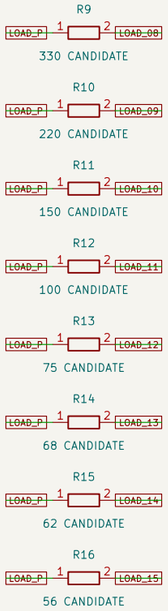
Sheet: /PowerMCU/
File: PowerMCU.kicad_sch

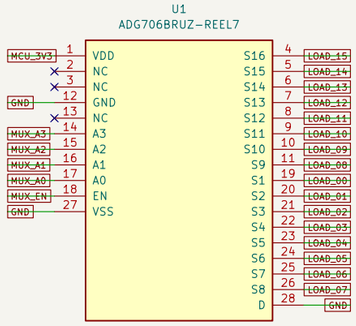
Title: Marc USB – PowerMCU – DRAFT NOT FOR FAB

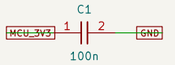
Size: A3	Date: 2026-10-18	**Rev: D0**
KiCad E.D.A. 10.0.5		Id: 2/7

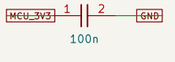


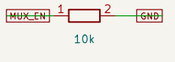


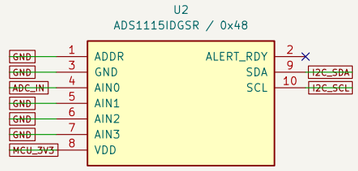


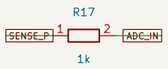


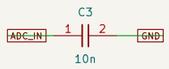
















ADG706 datasheet checked: 30mA continuous absolute max per channel; 0..VDD analog input range.

DSSC per-cell Voc 0.7V / Isc 34mA reported; confirm maxima under intended light and temperature.

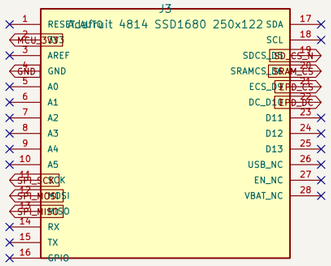
R1..R16 are independent nominal candidates, NOT recovered Marc physical BOM; new calibration required.

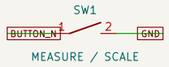
ADS1115: ADDR=GND => 0x48. GAIN_ONE +/-4.096V is NOT a 4.096V-safe input at VDD=3.3V.

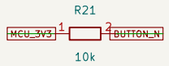
MUX_EN on GPIO14, 10k pulldown: boot disabled. Firmware patch required. No TPI MOSFET.

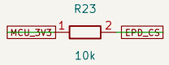
R17/C3 ADC filter only. K1 isolates DUT during harvest; R14-16 >=56R limit ADG706 current below 25mA at two-cell Voc 1.4V.

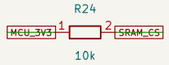
Sheet: /Measurement/	
File: Measurement.kicad_sch	
Title: Marc USB – Measurement – DRAFT NOT FOR FAB	
Size: A3	Date: 2026-10-18
KiCad E.D.A. 10.0.5	Id: 3/7











Adafruit 4814: SSD1680 / 250x122 / red–black–white; full Feather 16+12 mating pins.

3.3V ONLY at pin2. USB/VBAT/EN unused. Factory SJ1..SJ4 closed; SDCS pulled high; no SD.

Onboard auto-reset; RESET header pin1 NC. BUSY testpad not wired. MCU GPIO10/GPIO7 NC.

SW1 short press = scan; long press = scale menu; serial/web scale configuration patch required.

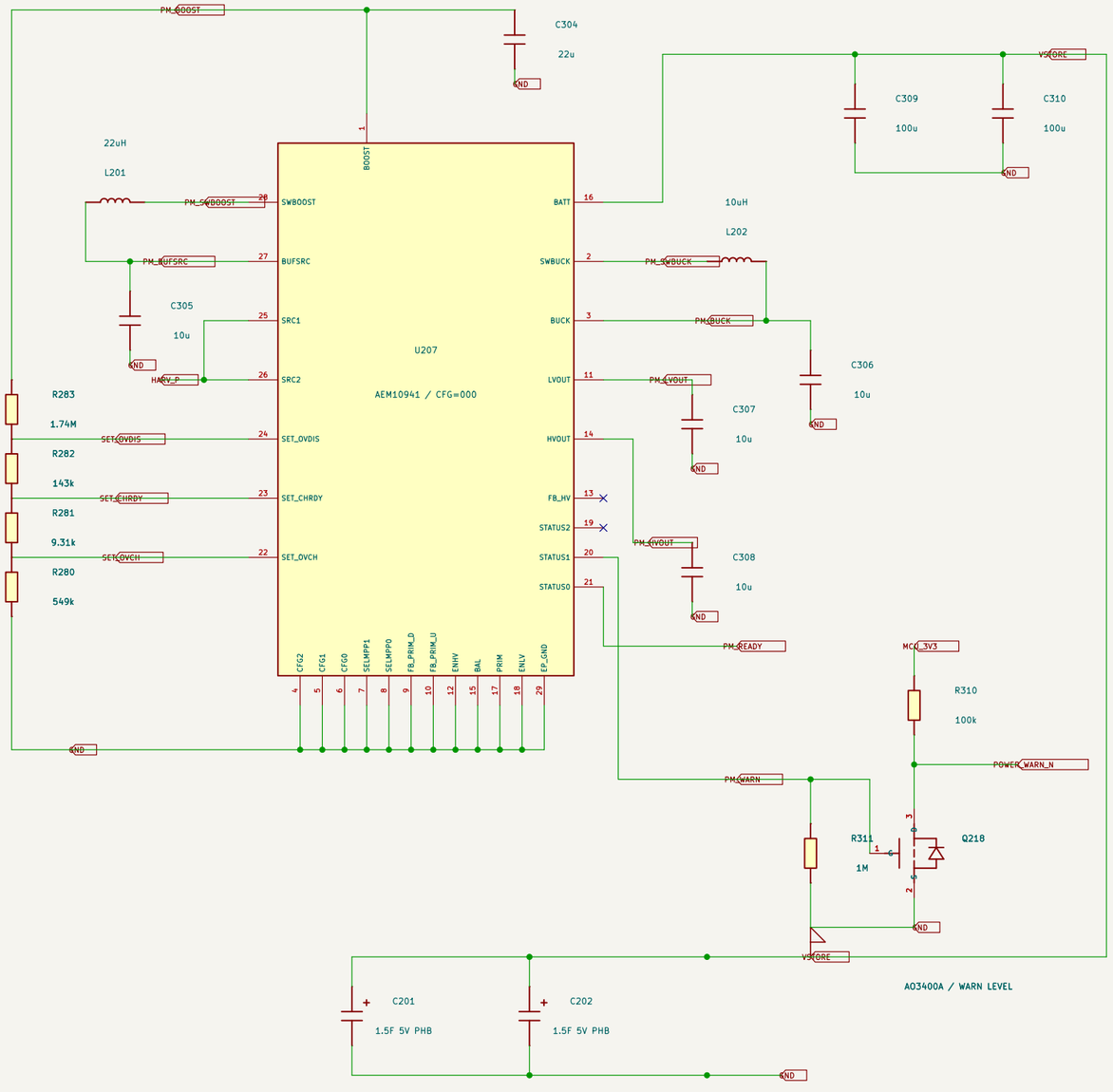
Collector A/B polarity, zebra material, force/resistance and plunger stack remain UNQUALIFIED.

Sheet: /DisplayUI/
File: DisplayUI.kicad_sch

Title: Marc USB – DisplayUI – DRAFT NOT FOR FAB

Size: A3	Date: 2026-10-18	**Rev: D0**
KiCad E.D.A. 10.0.5		Id: 4/7

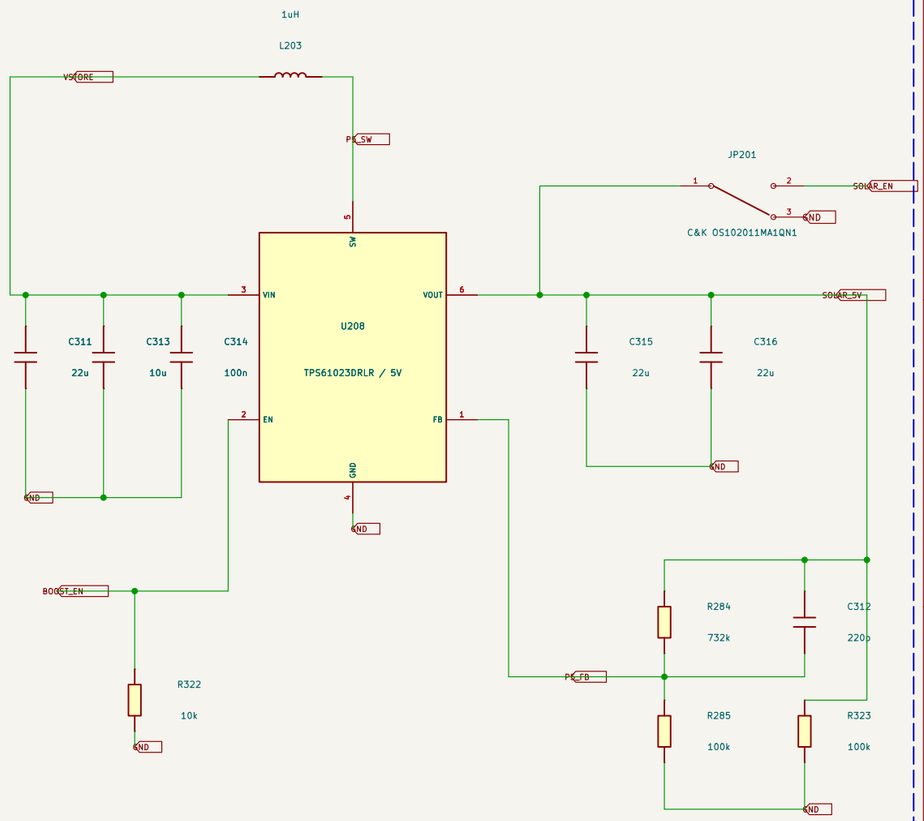
1. AEM10941 – HARVEST / CHARGER



C201/C202: two 1.5F PHB packs in PARALLEL: 3F / 5V total.

CFG=000; MPPT=70% Voc. Bottom configuration pins wired individually to GND.

2. TPS61023 – CONTROLLED 5V



R284/R285/C312: feedback group. R322: EN bias; R323: output bleed.

JP201 contacts carry only U214 ON control current; OFF before USB. U214 QOD left open to avoid sinking USB VBUS. Verify USB hot-plug and startup on hardware before fabrication relea

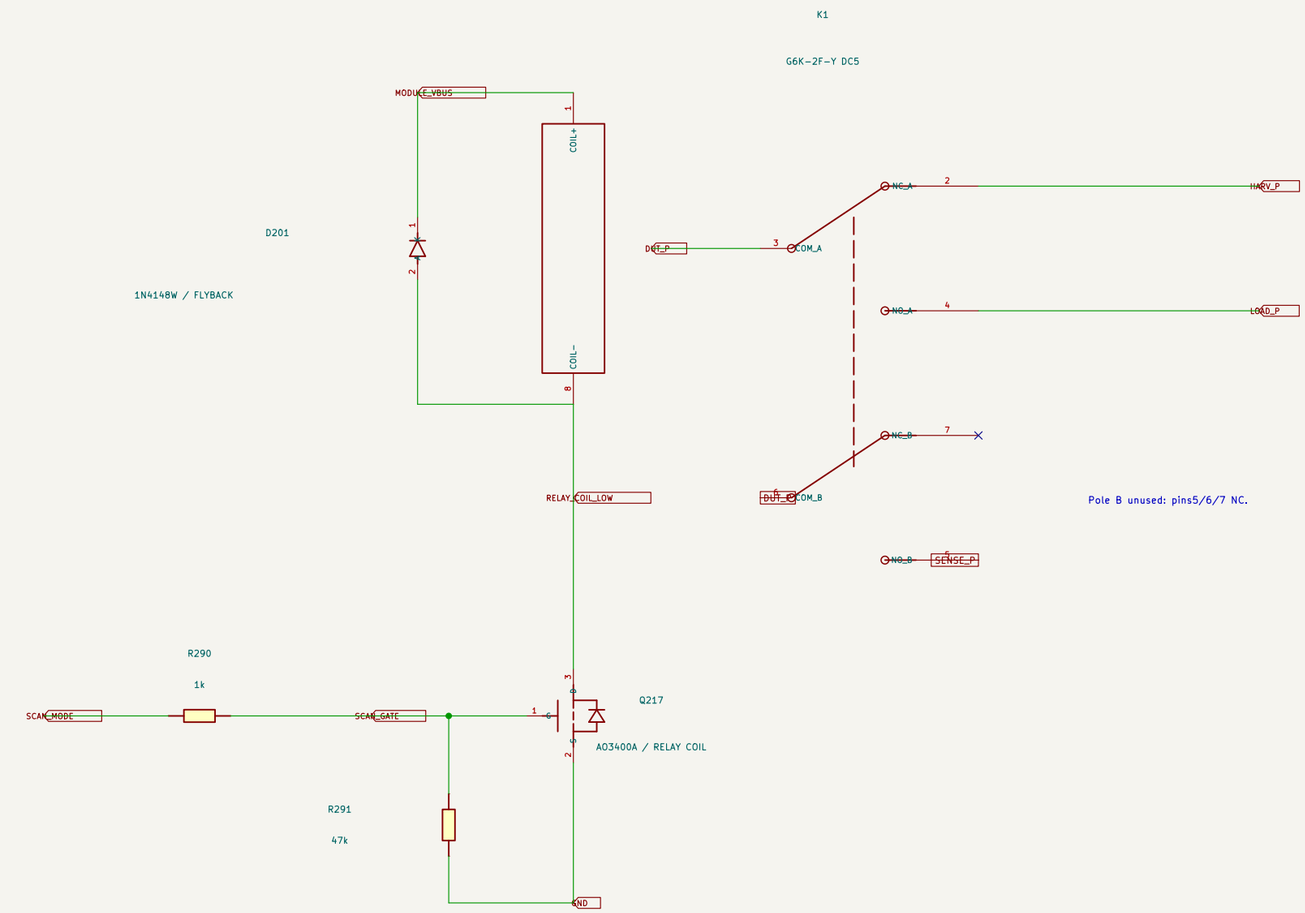
Solar Oracle Walkman

Sheet: /HarvestPower/
File: HarvestPower.kicad_sch

Title: POWER: AEM harvest + TPS61023 5V

Size: A3	Date:	**Rev: G20260913**
KiCad E.D.A. 10.0.5		Id: 5/7

K1 MECHANICAL RELAY — DEFAULT HARVEST / ENERGIZED MEASUREMENT



SCAN_MODE LOW / no power: COM3 -> NC2 -> HARV_P; LOAD_P disconnected.

SCAN_MODE HIGH: COM3 -> NO4 -> LOAD_P; HARV_P disconnected.

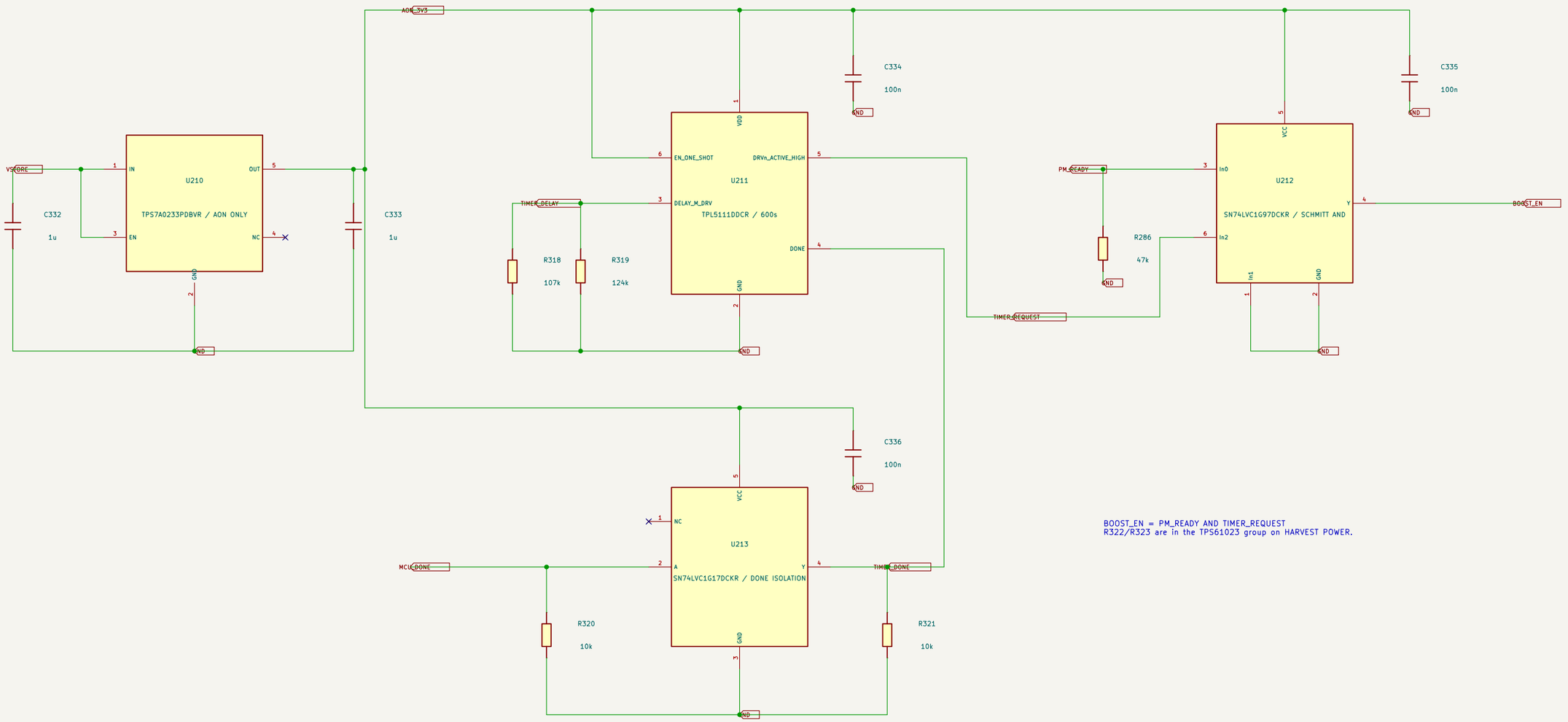
Coil: 5V / 237ohm / nominal21.1mA. Pin1 positive, pin8 low-side driver.

Disable 595 outputs and latch zero before switching. Wait for contact settling before sampling.

Flyback diode extends release time. Measure settling/release; 3ms relay spec is not a diode-clamped bound.

Solar Oracle Walkman		
Sheet: /HarvestSwitch/		
File: HarvestSwitch.kicad_sch		
Title: SWITCHING: K1 relay / coil driver / flyback		
Size: A3	Date:	**Rev: K1-RELAY**
KiCad E.D.A. 10.0.5		Id: 6/7

3. ALWAYS_ON — SUPPLY / TIMER / READY / DONE



BOOST_EN = PM_READY AND TIMER_REQUEST
R322/R323 are in the TPS61023 group on HARVEST POWER.

U210 + C332/C333: AON supply. U211 + C334/R318/R319: timer.

U212 + C335/R286: READY permission. U213 + C336/R320/R321: DONE interface.

Solar Oracle Walkman

Sheet: /AlwaysOn/
File: AlwaysOn.kicad_sch

Title: ALWAYS_ON: READY / timer / DONE

Size: A3	Date:	**Rev: G20260913**
KiCad E.D.A. 10.0.5		Id: 7/7